Renal tumor

- Benign
- Malignant

Benign (usually) adult tumors

- Renal papillary adenoma
- Angiomyolipoma
- Oncocytoma

Benign (usually) adult tumors

Renal papillary adenoma:

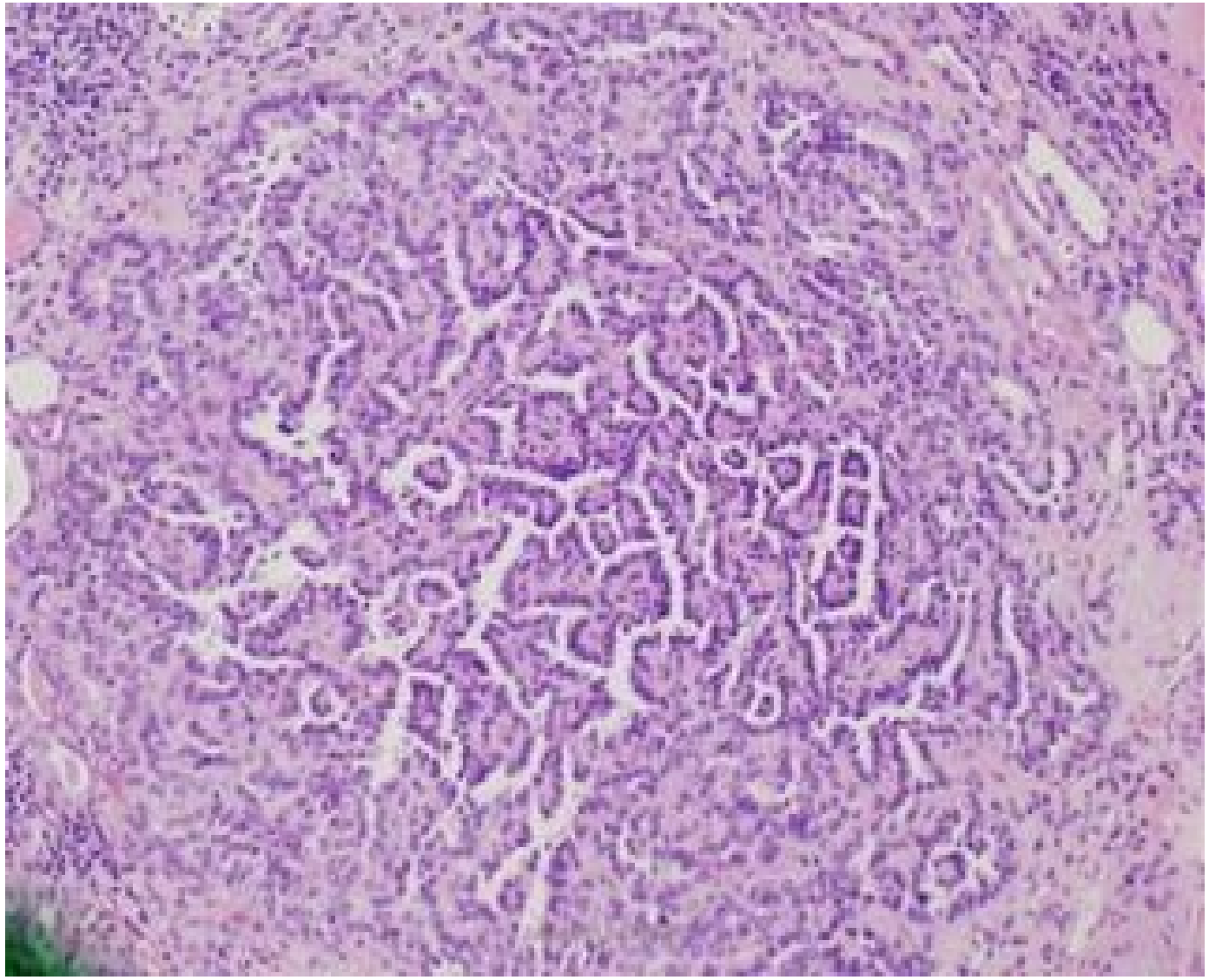
- Histogenesis: Renal tubular epithelium
- Morphology:
  - Small <0.5 cm [>1cm indicative of malignancy]
  - Well circumscribed
  - Pale yellow
  - Cortical tumors

NOTE: Irrespective of size all adenomas should be regarded as potentially malignant

Gross.

M/F/ Dense papillary, tubular, tubulopapillary of low cuboidal cells .





Angiomyolipoma:

- Benign
- 25-50% of patients with tuberous sclerosis caused by mutation in TSC1 or TSC2 tumor suppressor gene (Cerebral cortex lesion, skin abnormalities, epilepsy, intellectual disability and benign tumor in other sites ex. Heart.
- Increase susceptibility to spontaneous haemorrhage because of large size

Gross appearance of Angiomyolipoma

Well-circumscribed AML with a yellow-shite cut surface

Friable AML confined i
kidney parenchyma

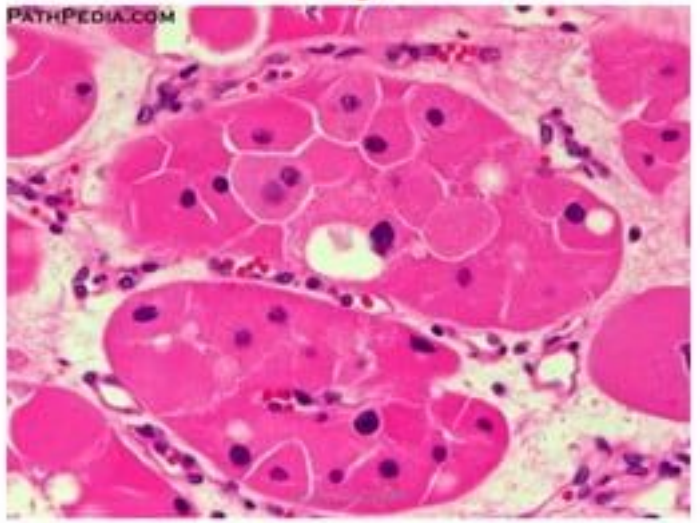
From: Flum AS, et al. J Urol. 2016 Apr;195(4 Pt 1):834-46.

Classic angiomyolipoma. (A) The tumor is separated from the kidney parenchyma and has a fat tissue-like appearance. (B) Histologically, the tumor contains fat tissue (red asterisks), smooth muscle cells (black arrow), and thick blood vessels with hyaline walls (black asterisks). The image has a magnification factor of ×100.

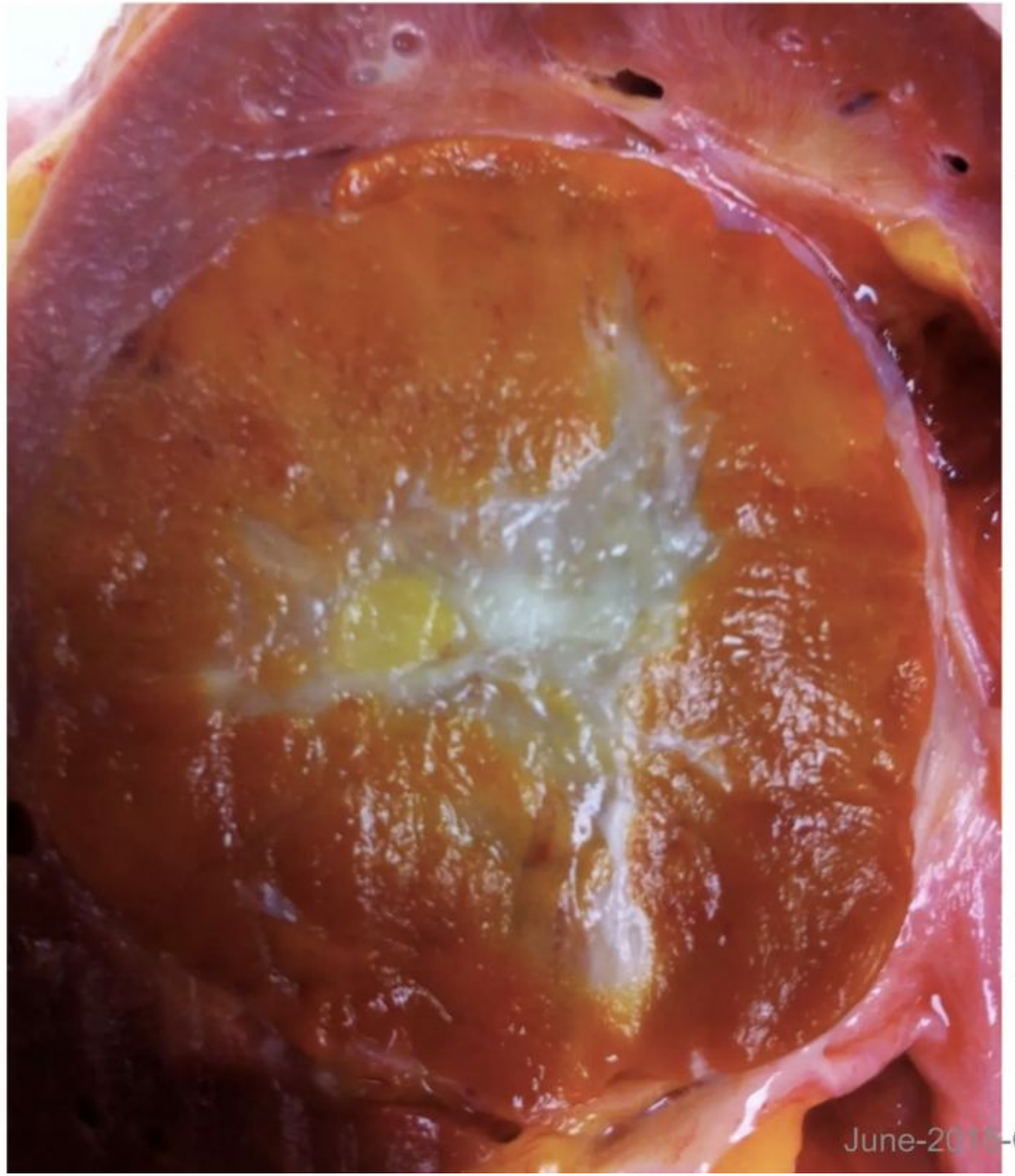
Oncocytoma

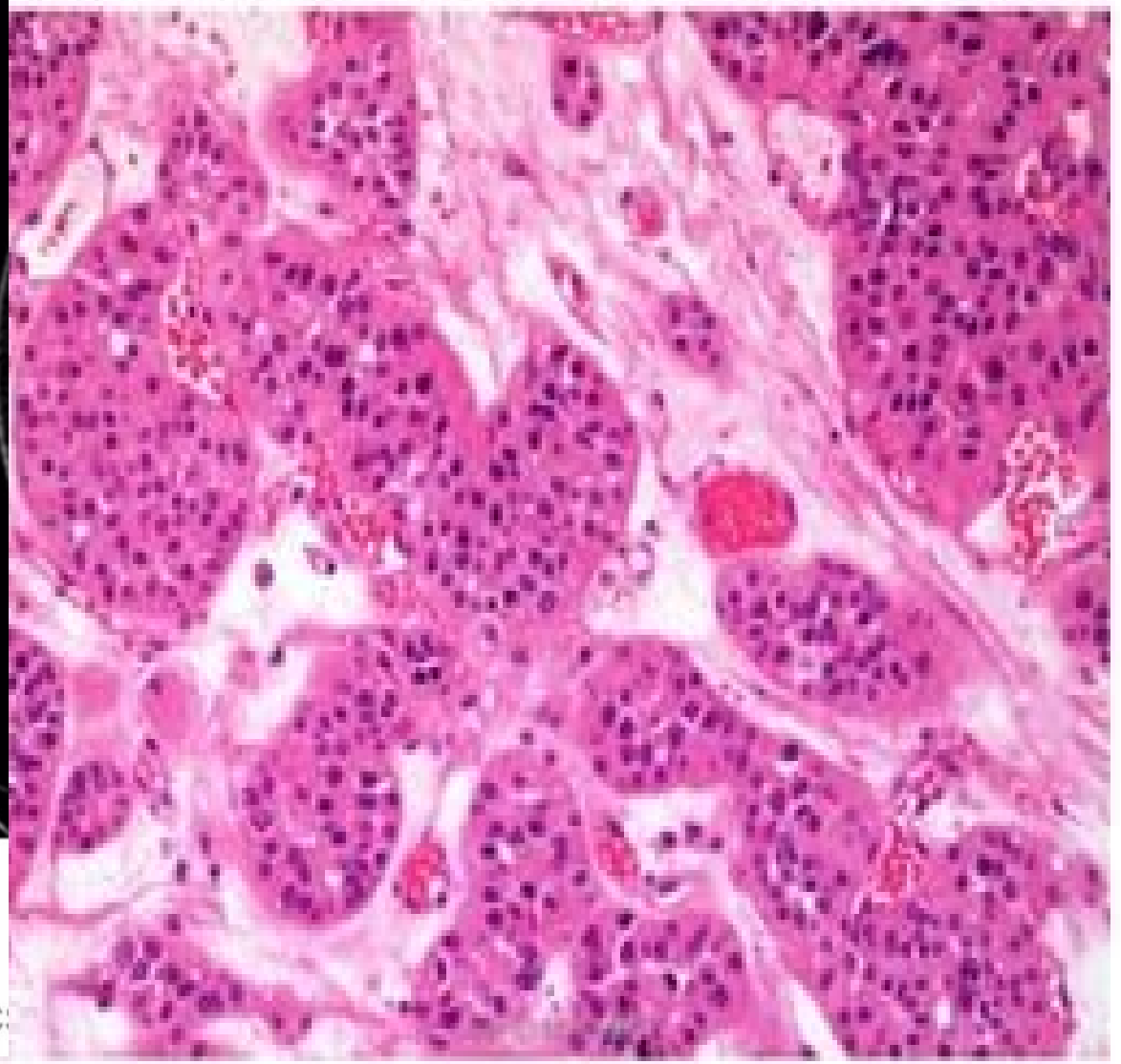
- Histogenesis: Intercalated cells of collecting duct
- Gross:
  - Tan / Mahogany brown
  - Well encapsulated
  - Central stellate scar
- Histology:
  - Tumor composed of large eosinophilic cells
- EM:
  - Numerous mitochondria

Oncocytoma



Oncocytoma





Malignant tumor

- **Blood in the urine (hematuria)**
- **Pain in the side or back that doesn’t go away**
- **Lump or mass in the side or abdomen**
- **Fatigue**
- **Unexplained weight loss**
- **Loss of appetite**
- **Fever**

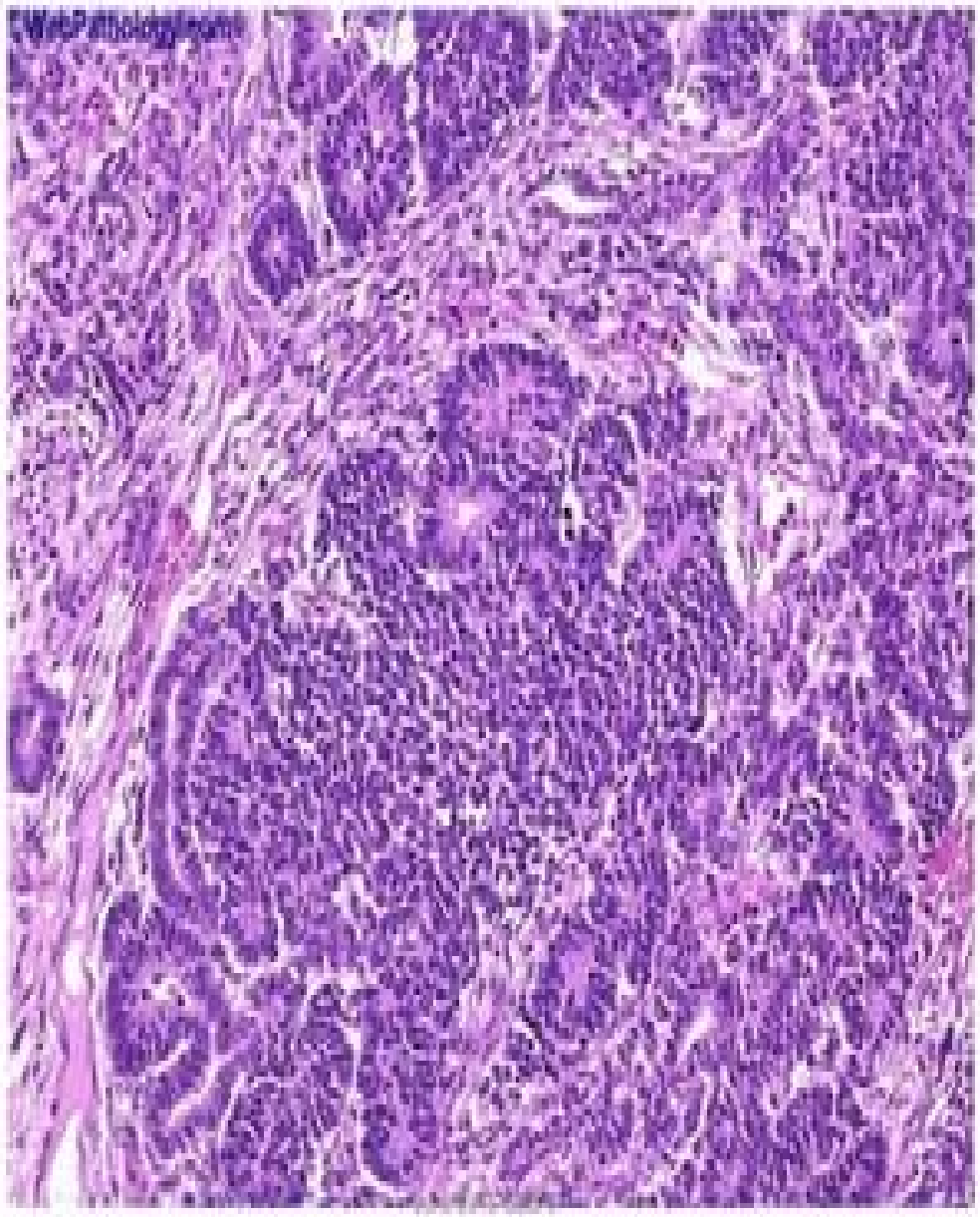
RCC; Secondary polycythemia (erythropoietin production).

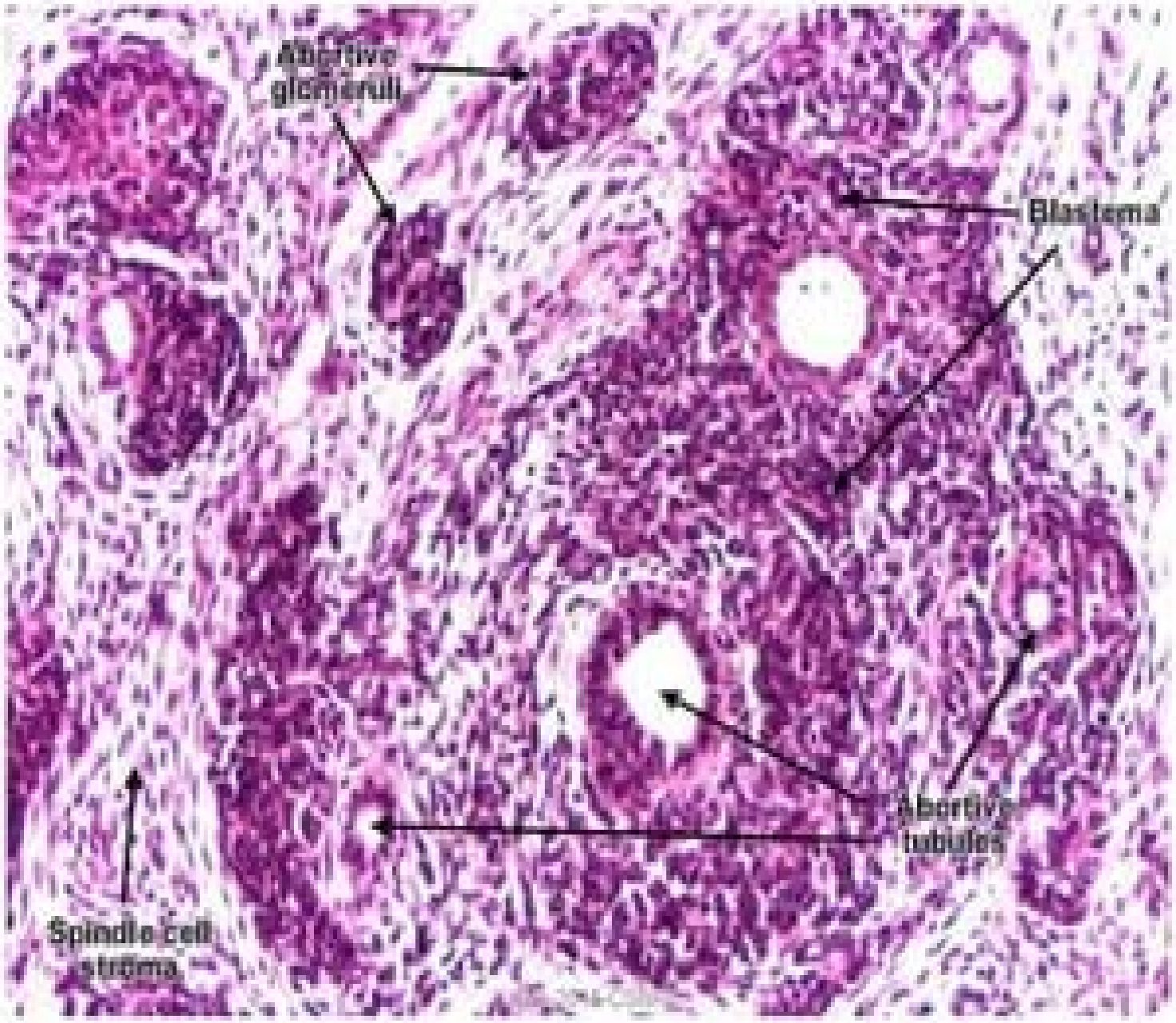
Ectopic production of various hormones or hormone-like substances (ACTH, prolactin, gonadotropins, renin, and parathyroid-like hormone) which can cause Paraneoplastic syndrome.

Wilms tumor

- Nephroblastoma (or) Wilms tumor is a malignant embryonal tumor. Thought to develop from persistent metanephric tissue or nephrogenic rests
  - *WT1* gene at chromosome 11p13 implicated in tumorigenesis
  - *WT2* gene at chromosome 11p15 (still not isolated) is a second locus
- (triphasic) Tumor consists of
  - ❖blastemal,
  - ❖epithelial and
  - ❖stromal elements
- Slight female predominance
- Most common: 3 - 4 years of age

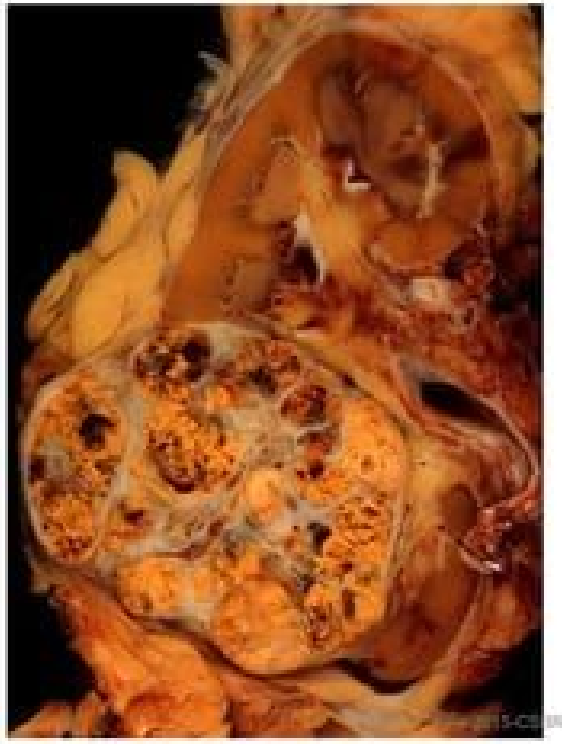





Abortive glomeruli
Blastema
Abortive tubules
Spindle cell stroma

Renal cell carcinoma

- Adult
- 2;1 male predilection
- Risk factors;
- ✓ Smoking, asbestosis
- ✓ Obesity
- ✓ Hypertension
- ✓ Drugs
- ✓ ESRD, chronic kidney disease, acquired cystic disease



Renal cell carcinoma

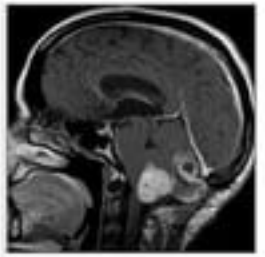
Epidemiology

- ~ Most RCC ….sporadic
- VHL syndrome (sporadic and familial)
- Hereditary leiomyomatosis & RCC…. AD
- Hereditary papillary carcinoma ..AD, multiple bilateral tumor with papillary morphology
- Birt-Hogg-Dube syndrome
- A rare variant occurring most often in young patients is associated with chromosomal translocations involving the TFE3 gene located at Xp11.2 locus

Von Hippel–Lindau disease



- Angiomatosis
- Hemangioblastoma
- Pheochromocytoma
- Papillary cystadenoma of epidydimis / broad ligament
- Endolymphatic sac tumor
- RCC



Classification of RCC; Histology, cytogenetic

(1) Clear cell RCC:

Most common

Any portion of kidney but commonly poles

Loss of sequences on the short arm chromosome 3

Usually solitary unilateral lesion

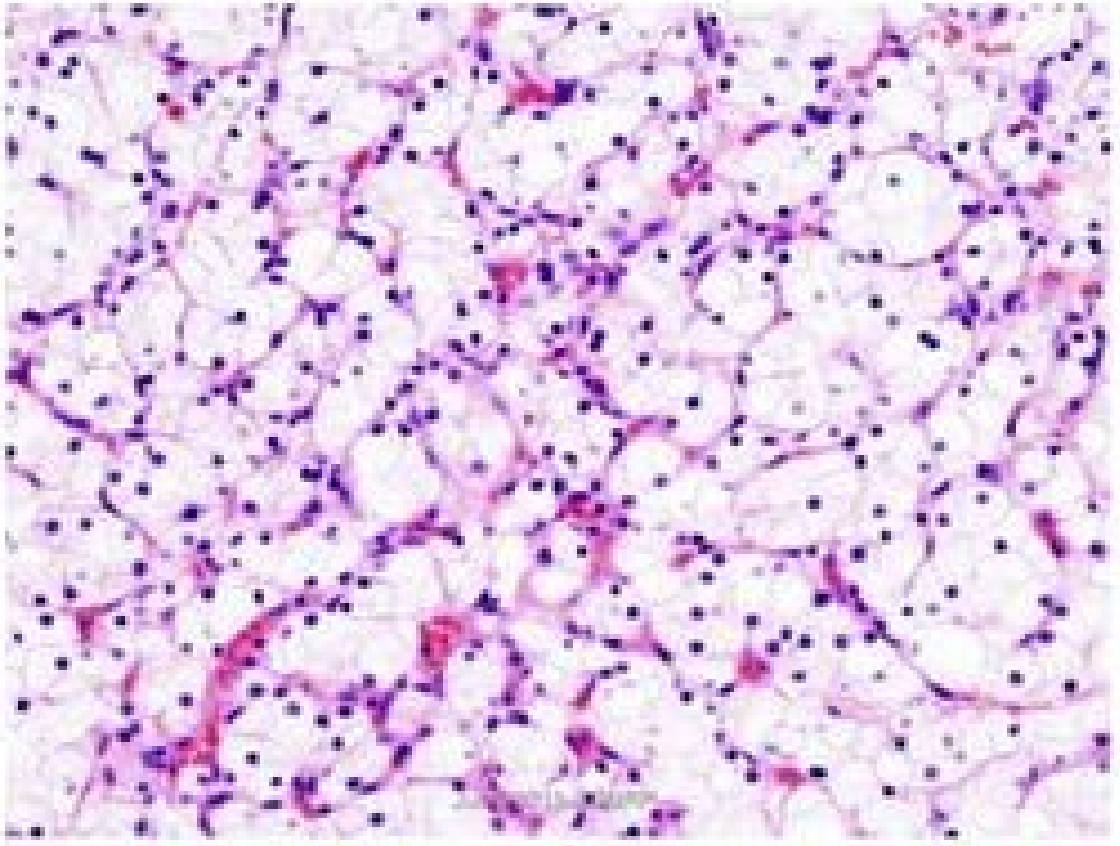
Frequently invades the renal veins or inferior vena cava some times as far as the of the right side heart

Renal cell carcinoma



Bright yellow-grey whit spherical mass of variable size

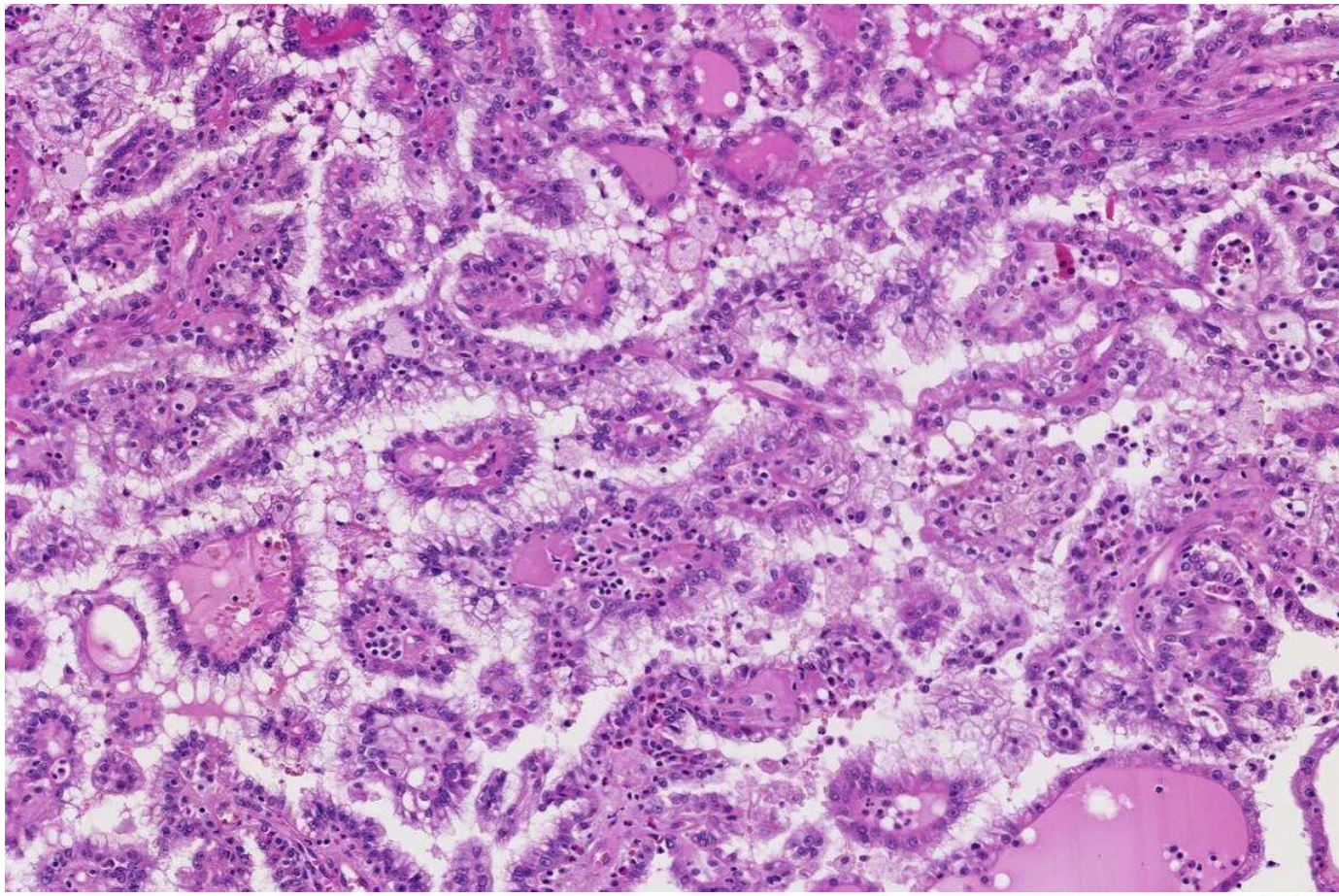
Renal cell carcinoma - Clear cell type

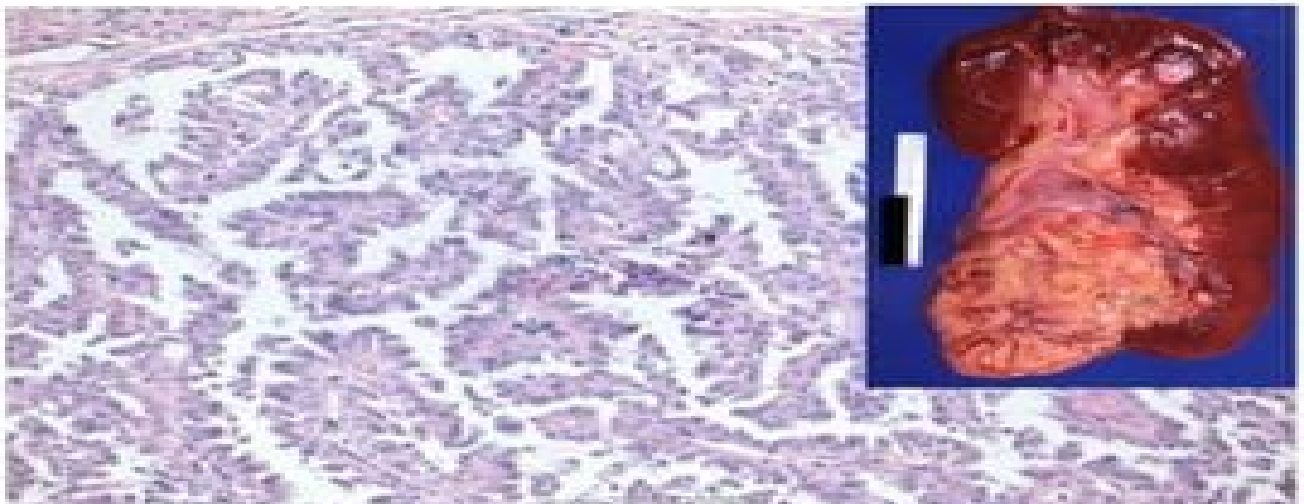


Round or polygonal with abundant clear cytoplasm which contain glycogen and lipid

(2) Papillary RCC:

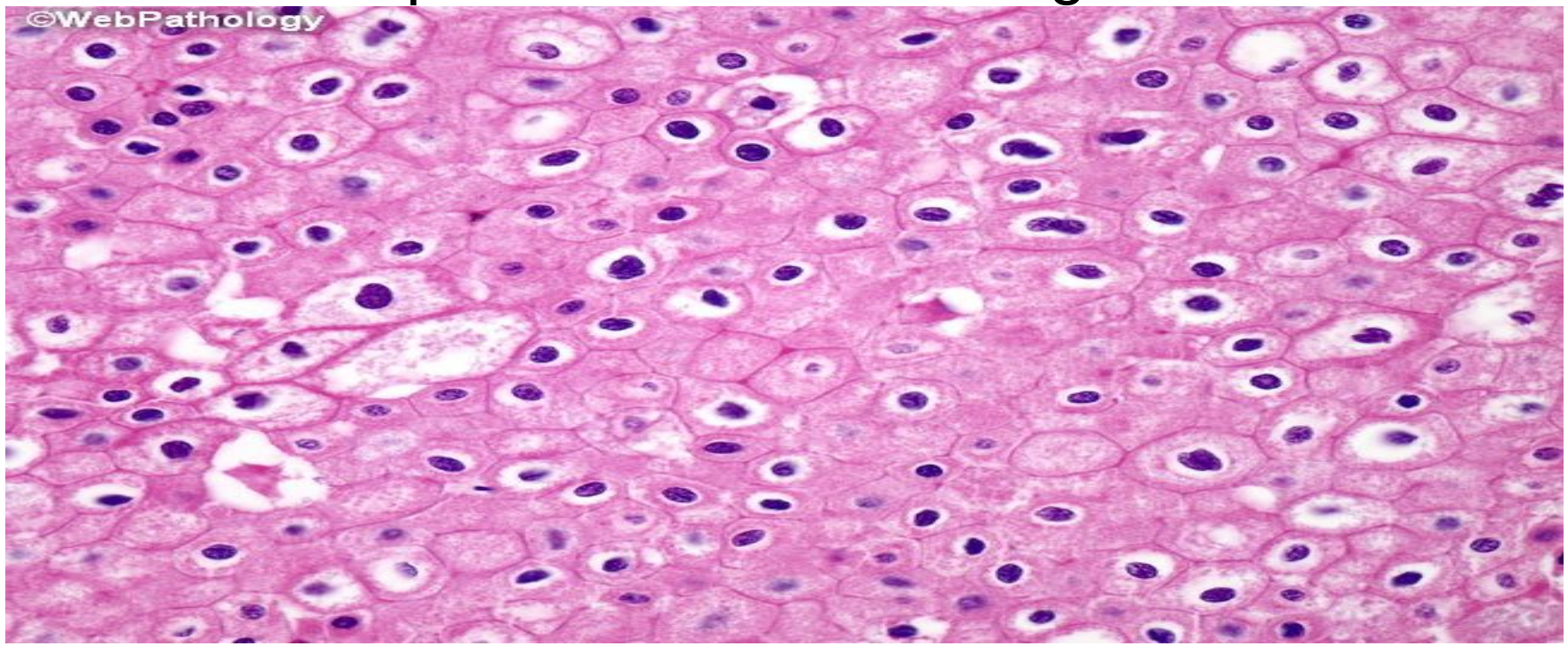
- Can be multifocal and bilateral
- 3p deletion





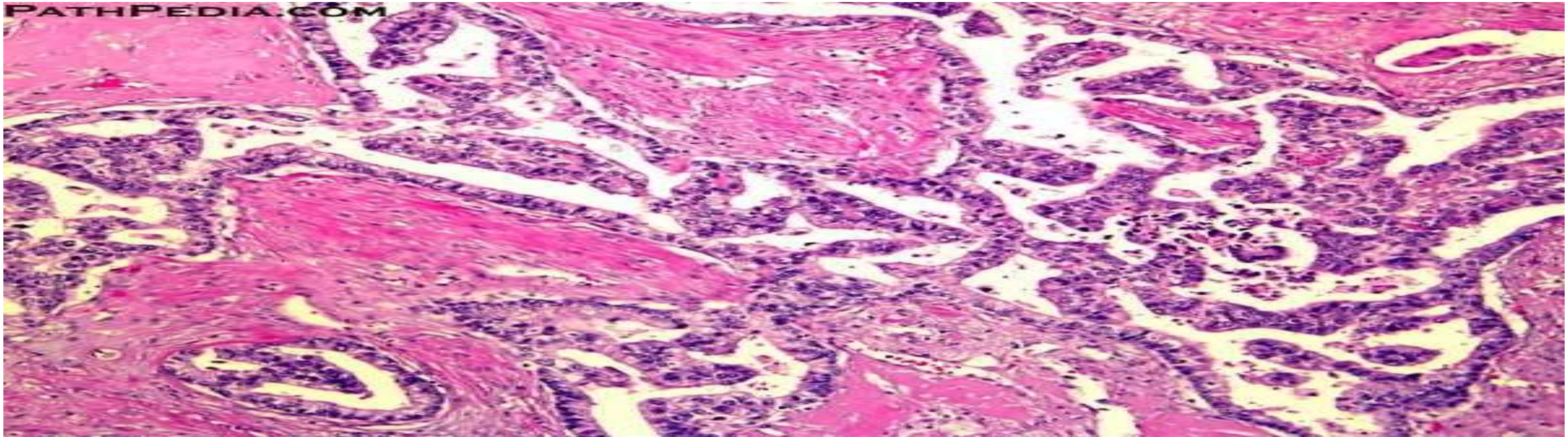
(3) Chromophobe RCC:

- Closely mimic oncocytoma, and like oncocytoma arise from intercalated cells of collecting ducts.
- Excellent prognosis.
- Microscope: eosinophilic granular cells with prominent cell borders with a perinuclear halo arranged in solid sheets.



(4) Collecting duct (Bellini duct) RCC:

- <1%
- Collecting duct cells in medulla
- Microscope: Irregular channels lined by highly atypical epithelium with a hobnail pattern within a prominent fibrotic stroma.



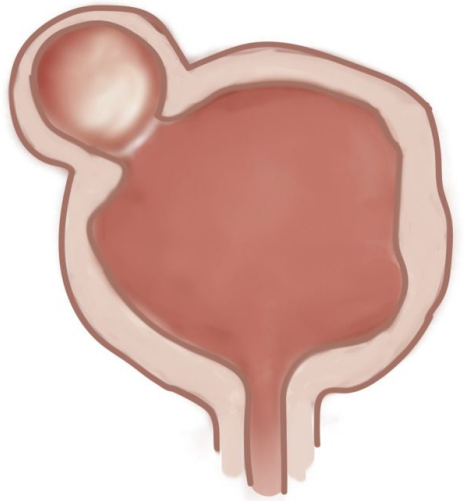
Urinary Bladder:

- Congenital anomalies
- Inflammation
- Tumor

Diverticula:

- Round to ovoid sac like pouch.
- May be congenital, but more commonly acquired.
  - **Congenital diverticula**: due to failure of development of normal musculature, or urinary tract obstruction during fetal development.
  - **Acquired diverticula:** due to prostatic enlargement (hyperplasia or neoplasia).

Bladder Diverticulum



- Most are small and asymptomatic.
- May be clinically significant:
  - Urinary stasis: predispose to infection and formation of bladder calculi.
  - Vesicoureteral reflux: as a result of impingement on ureter.
  - Rarely, carcinomas may arise in bladder diverticula.

Exstrophy:

- A developmental failure in anterior wall of abdomen and the bladder.
- The bladder communicates directly with the abdominal surface
- The exposed bladder mucosa develops infections, and may undergo colonic glandular metaplasia.
- Exstrophy is associated with an increased risk of bladder adenocarcinoma later.



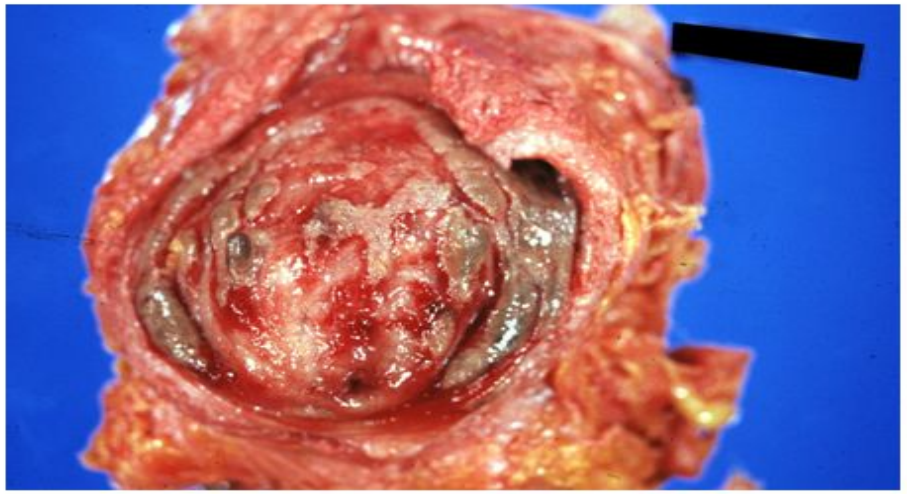
Urachal anomalies

- It’s a fibrous remnant of allantois (a canal that drains the urinary bladder of fetus that joins and runs within the umbilical cord), normally disappears before birth.
- May remain patent : total or partial.
  - Total patency: fistula between bladder and umbilicus.
  - Partial patency: urachal cysts lined by transitional or metaplastic epithelium; Carcinomas (mostly adenocarcinoma) may arise in such cysts.

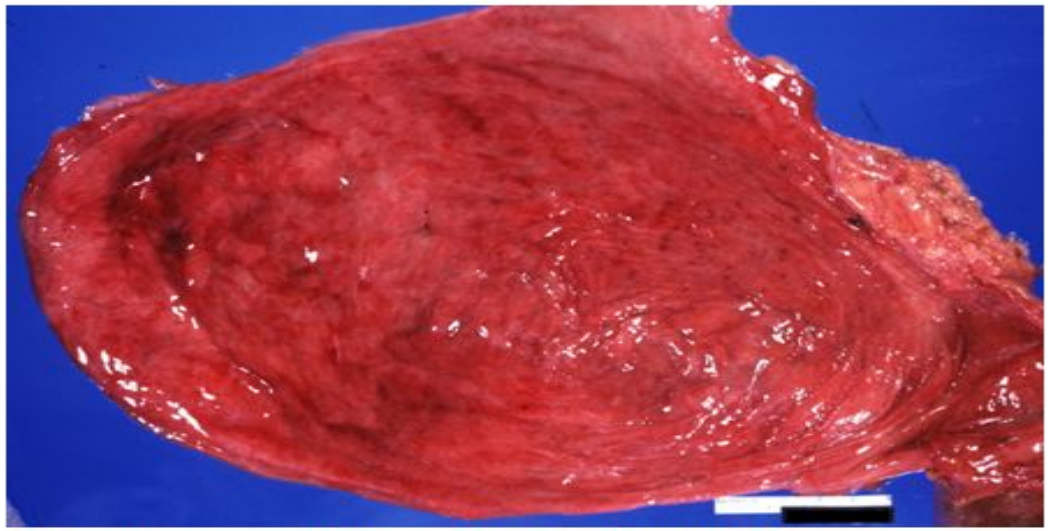
Inflammations:
Acute and Chronic Cystitis:

- Women are more likely to develop cystitis as a result of their shorter urethra.
- Etiologic agents: Bacteria (E.coli, Proteus),T.B cystitis, Fungi (Monilia), Schistosomiasis, Viruses (adenovirus), Chlamydia, and Mycoplasma.
- Predisposing factors: Bladder calculi and urinary obstruction, Diabetes mellitus, Instrumentation, Immune deficiency, Chemotherapy, and Radiation.
- Clinically: triad of: frequency, suprapubic pain, and dysuria.

- **Morphology: (acute and chronic cystitis)**
  - **Hyperemia of mucosa.**
  - **Exudate : Suppurative (purulent) cystitis.**
  - **Hemorrhage : Hemorrhagic cystitis**
  - **Cystitis follows radiation or chemotherapy : Epithelial atypia.**
  - **Chronic cystitis differs from acute in character of inflammatory infiltrate and fibrous thickening of bladder wall.**



GROSS: Urinary Bladder, Cystitis: Ulcerative Purulent mucosal surface.



GROSS: Hemorrhagic Cystitis, opened bladder with diverticula.

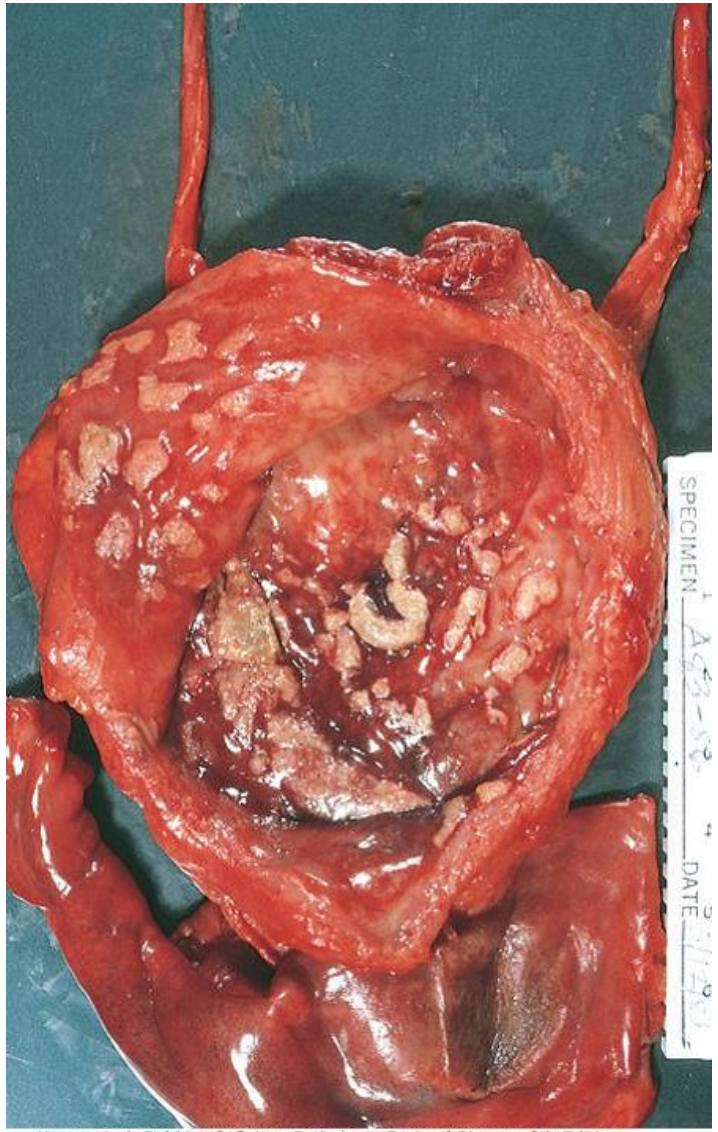
Special Forms of Cystitis:
Chronic pelvic pain syndrome

Interstitial Cystitis (Hunner Ulcer):

- Pain- pressure & discomfort
- Occurring most frequently in women.
- Inflammation and fibrosis of all layers of bladder wall.
- Cystoscopy findings: Fissures and punctate hemorrhages (glomerulations) in bladder mucosa which later on ulcerated.
- Main role of biopsy to rule out CIS
- Unknown etiology

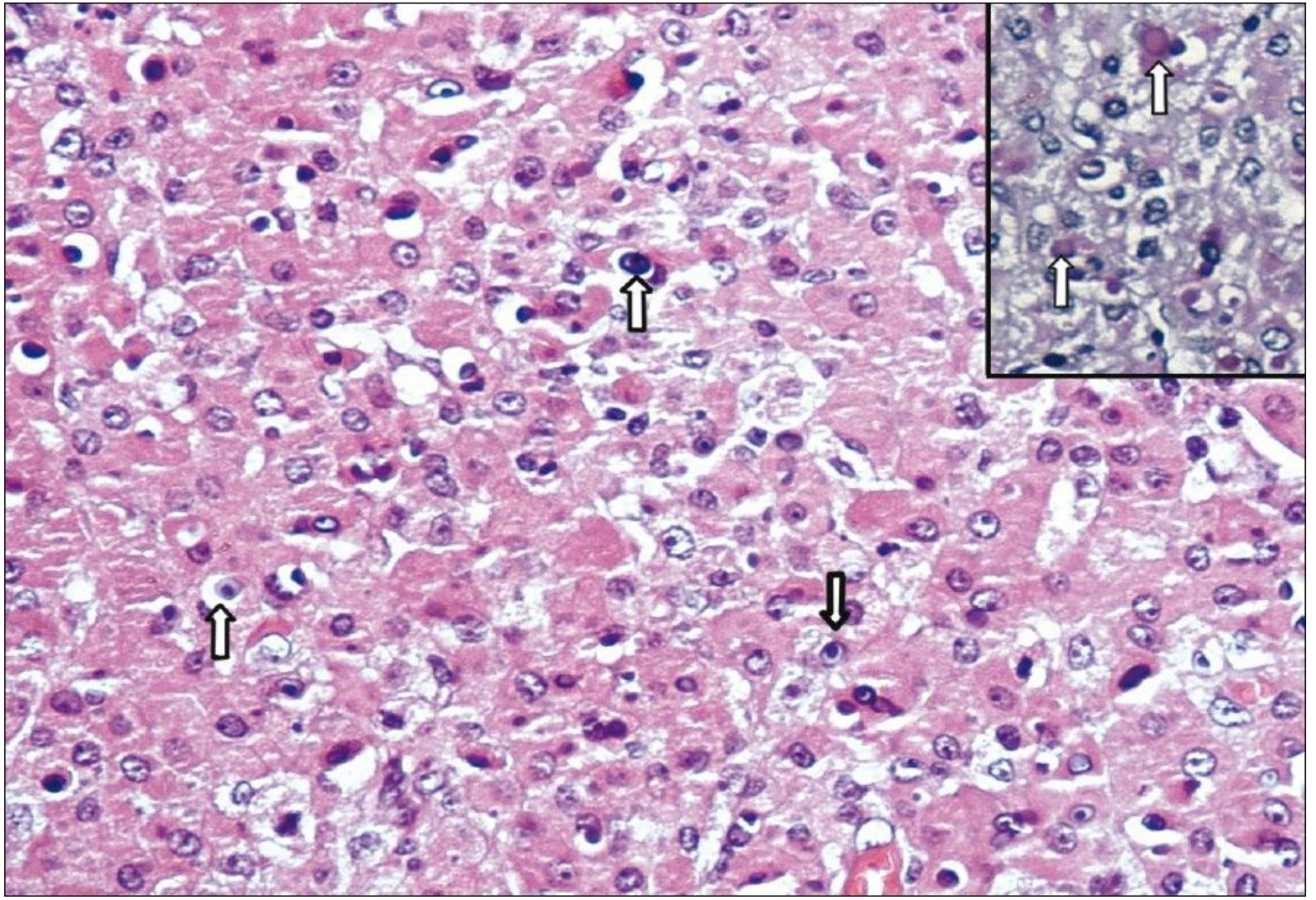
MalaKoplakia:

- Chronic inflammation
- Soft yellowish mucosal plaques 3-4cm.
- Related to chronic bacterial infection (E. coli or Proteus), frequency in setting of immunosuppressed or transplant recipients.
- Due to defects in phagocytic or degradative function of macrophages that become overloaded with undigested bacterial products.
- Microscope: infiltration of bladder mucosa by large macrophages with abundant granular cytoplasm **.**There is also laminated mineralized concretions resulting from deposition of calcium (**Michaelis-Gutmann bodies)** present both within the macrophages and in between.
- Similar lesions can occur in colon, lung, bone, kidney, prostate, and epididymis.



Kumar et al: Robbins & Cotran Pathologic Basis of Disease, 8th Edition.
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Malacoplakia (Gross): soft, yellow, mucosal plaques.



(H and E stain ×200): Numerous intracellular and extracellular Michaelis Gutmann bodies (arrows); Inset: Periodic acid Schiff stain (PAS stain).

Polypoid cystitis

- Irritation of bladder mucosa, most commonly as a result of instrumentation (catheter).
- Polypoid cystitis may be mistaken for papillary urothelial carcinoma

Urothelial tumors
WHO classification 2022

❖*Noninvasive urothelial tumors*

- Urothelial papilloma
- Inverted urothelial papilloma
- Papillary urothelial neoplasm of low malignant potential
- Noninvasive papillary urothelial carcinoma, low grade
- Noninvasive papillary urothelial carcinoma, high grade
- Urothelial carcinoma in situ

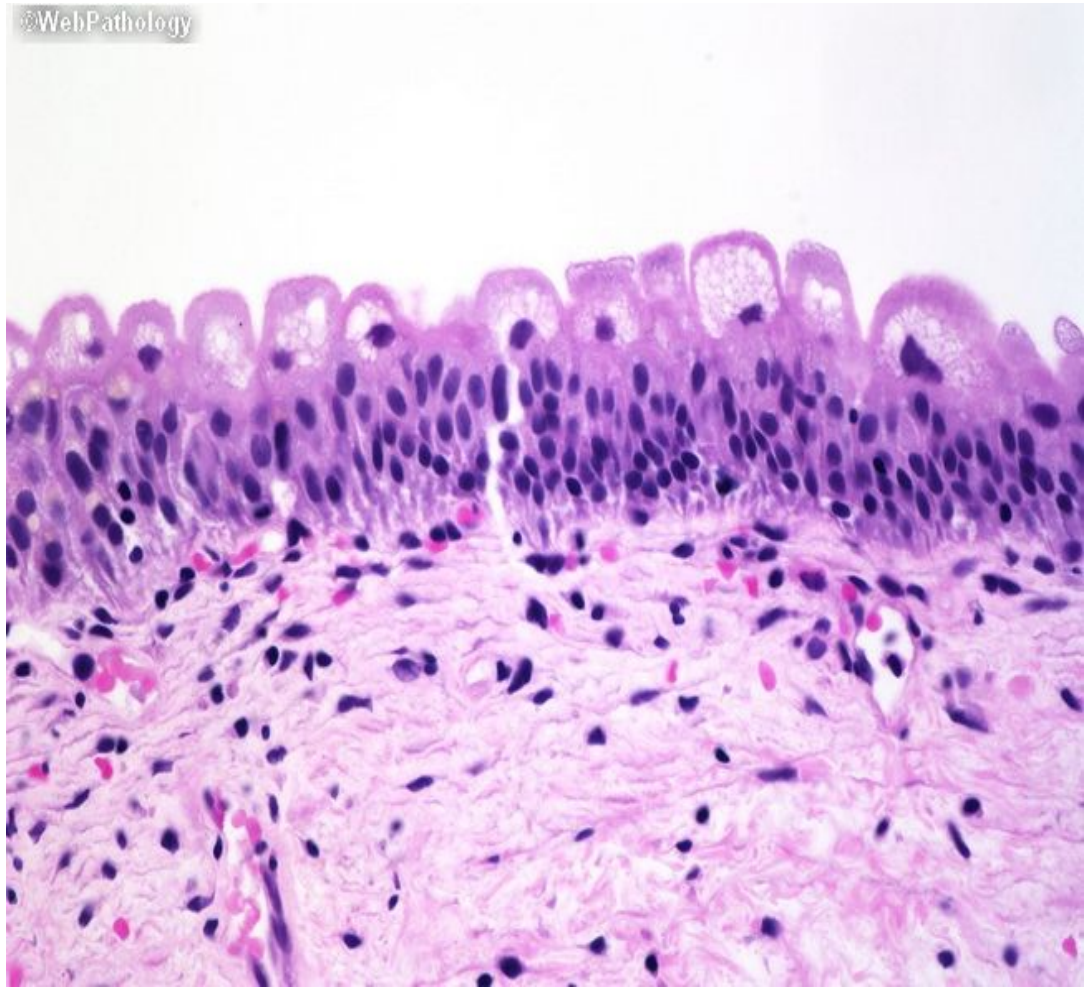
❖*Invasive urothelial neoplasms*

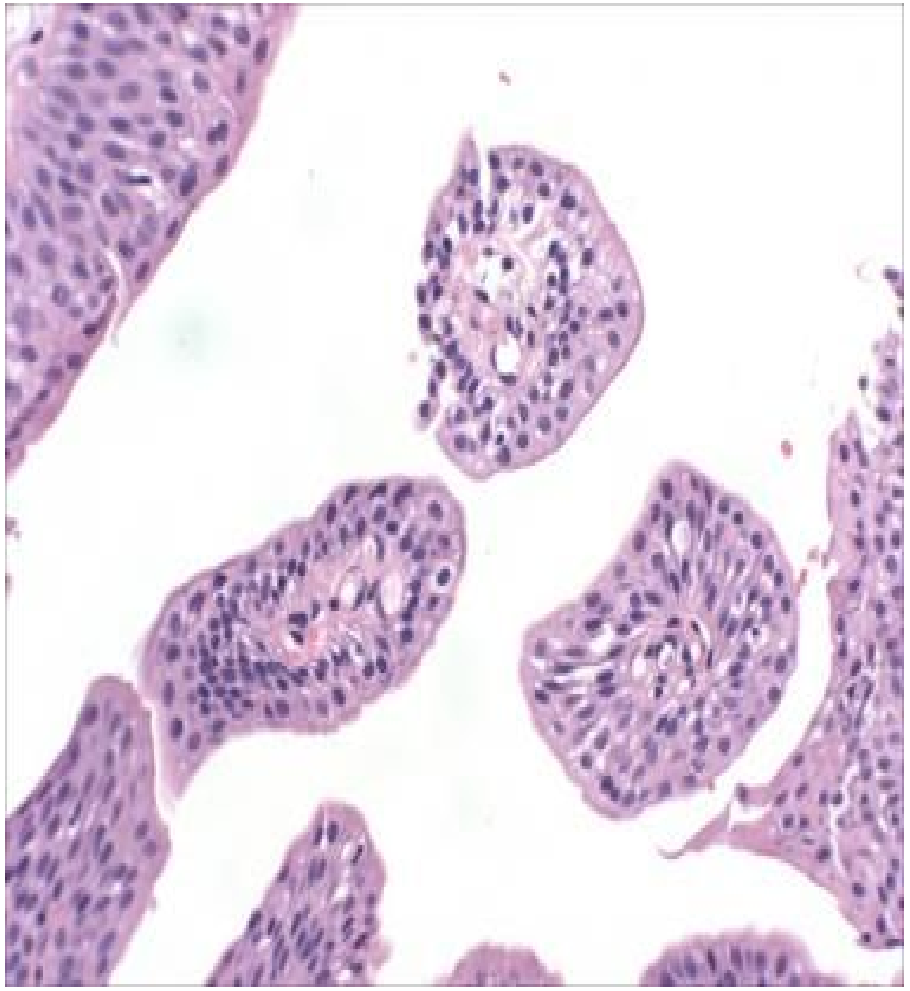
- Invasive urothelial carcinoma

Papilloma:

- Benign tumor, commonly seen in younger patients.
- Morphology: Finger -like papillae having fibrovascular core covered by transitional epithelial cells that are identical to normal urothelium (7- 8 cell layer thickness).
- Umbrella cell layer is present.

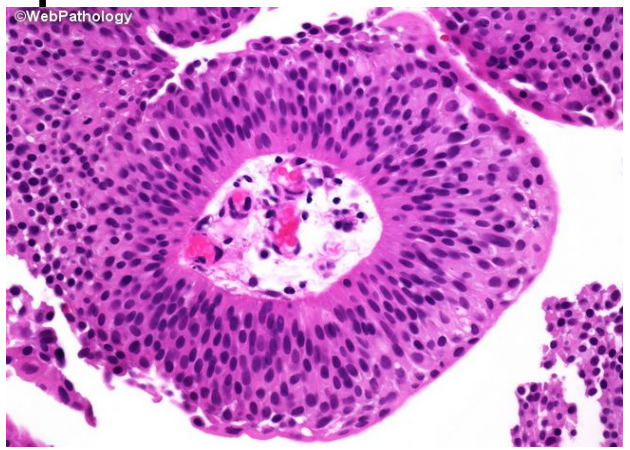
Urothelial papilloma: Fibrovascular core covered by urothelial cells.





Papillary urothelial neoplasms of low malignant potential (PUNLMP):

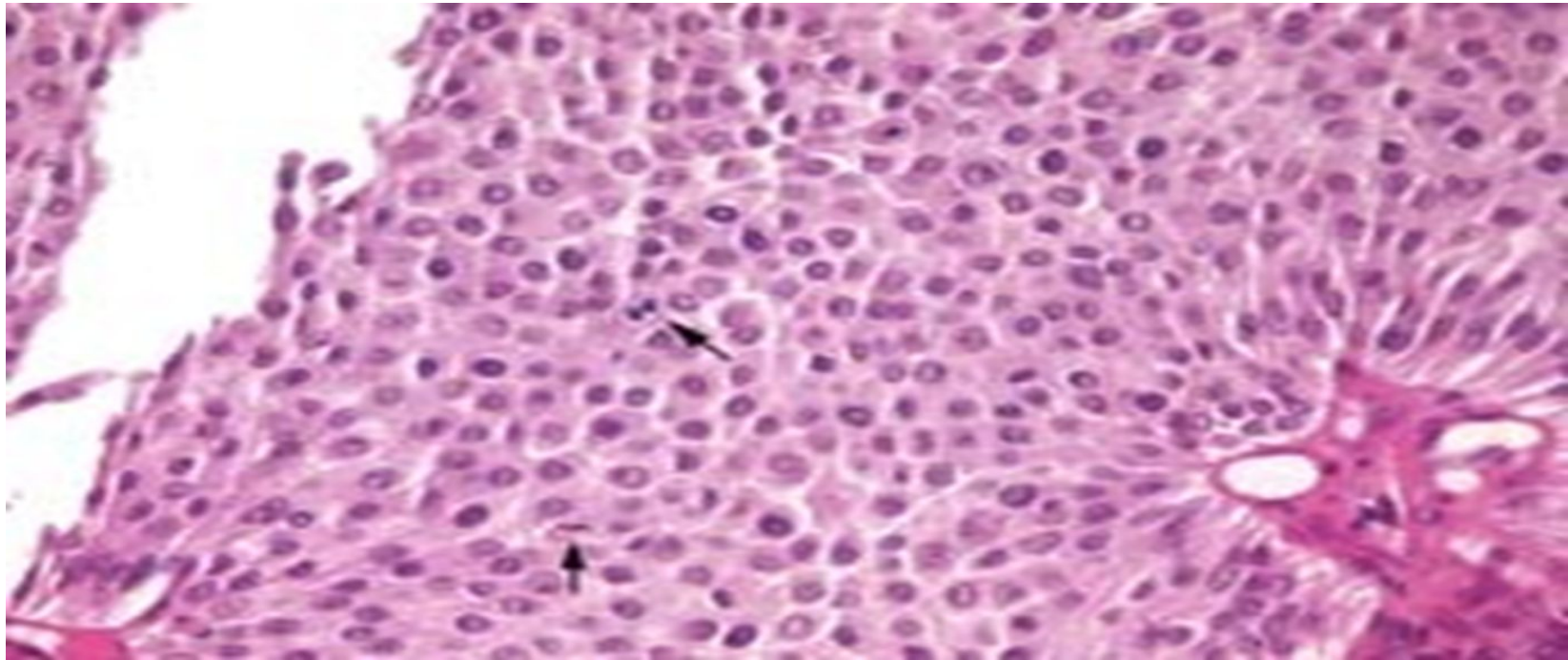
- The only difference from papilloma being thicker urothelium (> 8 cell layer thickness) .
- Mitotic figures are not seen (when present they are confined to basal layer).
- Not associated with invasion.
- Umbrella cell layer may present or absent.





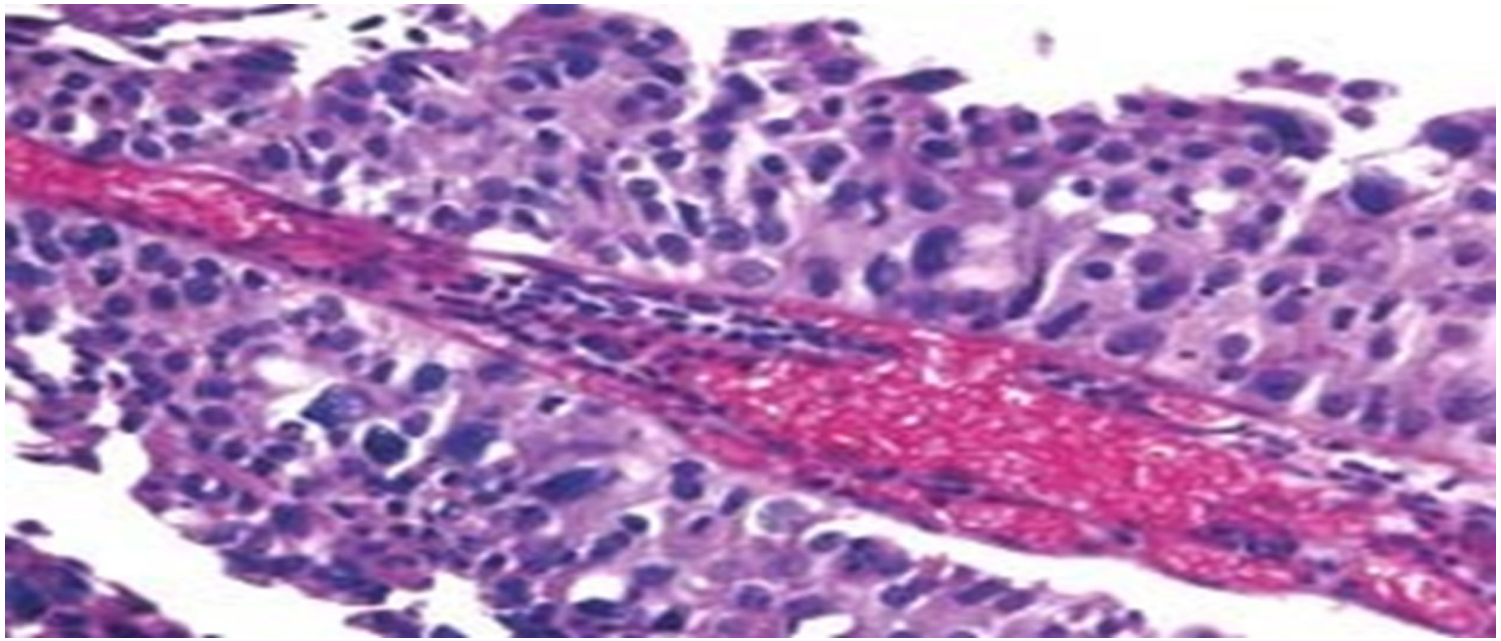
Low-grade papillary urothelial carcinoma:

- The thickness is more ; but the tumor cells still maintain polarity (are cohesive), and show mild to moderate atypia (scattered hyperchromatic nuclei, infrequent mitotic figures, and mild to moderate nuclear pleomorphism).
- Loss of umbrella cell layer.
- The tumor can recur, and can invade.



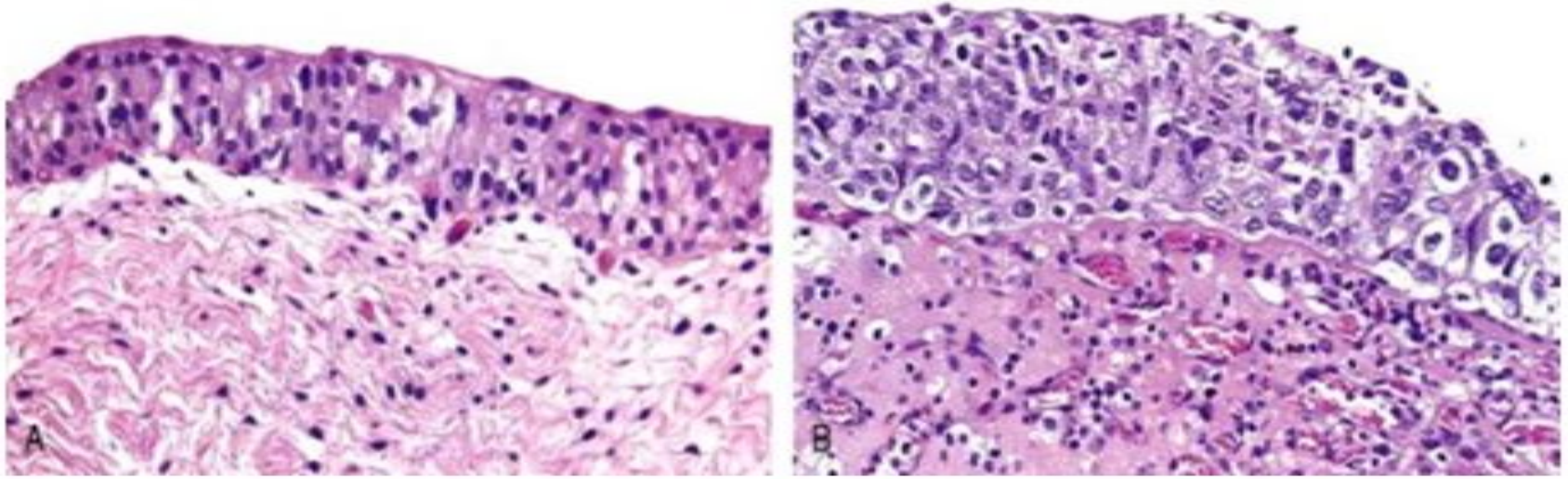
High-grade papillary urothelial carcinoma:

- The tumor cells are discohesive with loss of polarity, and show sever atypia (large hyperchromatic nuclei, frequent mitotic figures, marked pleomorphism, and necrosis).
- Loss of umbrella cell layer.
- Higher incidence of invasion into muscular layer, and significant metastatic potential.



Flat lesion:
Carcinoma in situ (CIS):

- The presence of malignant cells in flat urothelium and not invading the BM.
- Range from full-thickness to scattered malignant cells (pagetoid spread).



A, Normal urothelium

B, Carcinoma in situ

Invasive urothelial (transitional cell) cancer:
Extent of invasion (flat or papillary) is of prognostic significance.
Thus, staging (in addition to grading) is critical in assessment of bladder neoplasms.

Pathologic Staging of Bladder Carcinoma (T = PrimaryTumor): **AJCC/UICC: American Joint Commission on Cancer/Union Internationale Contre le Cancer.**

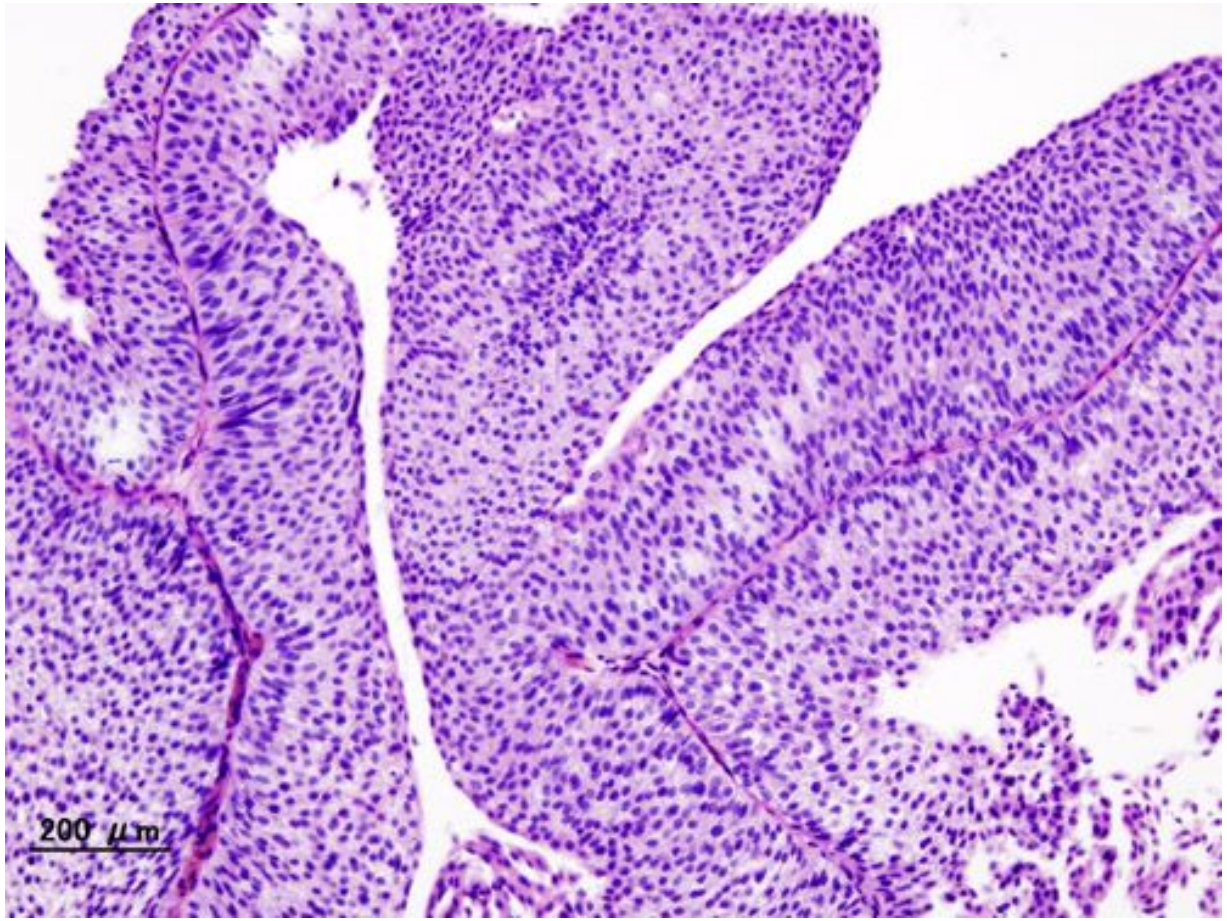
- **AJCC/UICC**
- **Noninvasive, papillary**
- **Carcinoma in situ (non invasive, flat)**

- **Lamina propria invasion**
- **Muscularis propria invasion**
- **Microscopic extravesical invasion**
- **Grossly apparent extravesical invasion**
- **Invades adjacent structures**

Depth of Invasion
Ta
Tis
T1
T2
T3a
T3b
T4

Epidemiology, clinical, and pathogenesis of urothelial (transitional cell) cancer:

- More common in male (ratio is 3:1).
- Most often the presenting feature is hematuria.
- Factors implicated in causation include:
  - Cigarette smoking.
  - Industrial exposure to naphthylamine (an aniline dye).
  - Long-term use of analgesics.
  - Long-term exposure to cyclophosphamide.
  - Prior exposure of bladder to radiation.



Other Types of bladder Carcinomas:

Squamous cell carcinoma (SCC):

- More frequent in countries endemic for urinary schistosomiasis, It can also be associated with renal calculi.

Adenocarcinoma:

- Arise from urachal remnants, or exstrophy.

Mesenchymal tumors in bladder:

- Benign: rare, the most common is leiomyoma.
- Malignant: Sarcomas:
  - The most common sarcoma in infancy or childhood is embryonal rhabdomyosarcoma.
  - The most common sarcoma in adults is leiomyosarcoma